EventForce Management System

Salesforce Implementation Report

Overview

The EventForce Management System is a cloud-based CRM solution engineered on Salesforce to digitize, centralize, and automate end-to-end operations for Event Planner Solutions. By replacing fragmented, manual spreadsheets, the CRM establishes an integrated platform handling client bookings, venue reservations, vendor service coordination, automated timeline alerts, and post-event feedback collection.

The system leverages declarative tools (Lightning App Builder, Record-Triggered Flows, Approval Processes, and Validation Rules) alongside programmatic enhancements (Apex Classes, Apex Triggers, and Batch/Schedulable Apex) to provide real-time operational visibility and eliminate human scheduling errors.

Objectives

Centralized Data Management: Unify disparate operational records (Clients, Events, Venues, Vendors, Feedback) into a single relational schema to ensure seamless cross-functional collaboration.

60% Reduction in Manual Processes: Automate multi-step approval workflows, timeline-based client notifications, venue status toggles, and daily past-event archival.

40% Improvement in Booking Efficiency: Eliminate scheduling conflicts and venue double-bookings via Apex trigger validation.

Enhanced Stakeholder Engagement: Send automated 3-day reminder notices to clients and coordinators to keep all parties aligned.

Centralized Data Management: Unify disparate operational records (Clients, Events, Venues, Vendors, Feedback) into a single relational schema to ensure seamless cross-functional collaboration.

Data-Driven Decision Making: Deliver real-time operational insights into event distribution, monthly budgets, and vendor performance via custom reports and dynamic dashboards.

Strict Data Governance and Security: Establish private baseline access across sensitive event data while leveraging role hierarchies, permission sets, and targeted sharing rules to guarantee appropriate visibility for administrators, coordinators, and external partners.

Standardized Financial Forecasting: Automate accurate event budget calculations across various event categories through dynamic formula logic, removing guesswork and manual cost entry errors.

Optimized Resource Allocation: Maintain real-time tracking of venue availability states and service partner assignments through custom junction relationships, preventing resource underutilization and overlapping vendor contracts.

Closed-Loop Quality Control: Automate client feedback collection and contextual validation to directly evaluate vendor reliability and overall event success, driving continuous service enhancement.

Operational Continuity and Scalability: Ensure zero system disruption during high-volume booking cycles by handling routine maintenance through scheduled background batch processes and scalable cloud infrastructure.

- **Centralized Data Management**: Unify disparate operational records (Clients, Events, Venues, Vendors, Feedback) into a single relational schema to ensure seamless cross-functional collaboration. **95% Billing Accuracy**: Auto-generate itemized invoice line items directly from confirmed bookings, space rates, and vendor add-ons.
- **Real-Time Operational Visibility**: Surface venue utilization rates, sales pipeline health, and revenue forecasts through centralized operational dashboards.

PHASE 1: REQUIREMENT ANALYSIS & PLANNING

Functional Scope: Design custom data architecture, enforce data hygiene via validation rules and lookup filters, automate operational updates via Salesforce Flows and Apex, establish cancellation governance, enforce granular role-based security, and implement executive reporting.

Data Governance & Integrity Constraints:

Enforced clean data entry at inception through contextual lookup filters specifically restricting feedback event lookups strictly to events associated with the selected client.

Designed automated budget formula logic to calculate base financial projections without manual calculation.

Formulated validation rules to safeguard essential customer contact data against improper formatting.

Stakeholder Mapping:

Clients: Request event services, receive schedule alerts, and submit feedback.

Event Coordinators: Manage booking schedules, track vendor deliverables, and initiate cancellation approvals.

Vendors: Deliver assigned operational services (catering, photography, decor, DJ/audio).

Venue Managers: Track physical space reservations and availability.

Event Administrators: Maintain system integrity, monitor batch jobs, and manage user security.

PHASE 2: BACKEND DEVELOPMENT & CONFIGURATIONS

Data Schema & Relationships

Custom Objects & Schema:

Client__c: Tracks client identity and regional demographics (Client Name, Email, Phone, Address, Country, City).

Venue__c: Manages physical event spaces (Venue Name, Address, Location, Capacity, Availability Status [Available, Reserved, Not Available]).

Vendor__c: Catalogs external service partners (Vendor Name, Email, Phone, Service Type [Multi-Select], Status).

Event__c: Core transaction object (Event Name, Event Date, Event Type, Event Status, Event Budget).

Feedback__c: Captures client satisfaction (Feedback ID [Auto-Number: F-{0000}], Rating, Comments).

EventVendor__c: Junction object establishing a Many-to-Many relationship between Event__c and Vendor__c via dual Master-Detail relationships.

Relational Model:

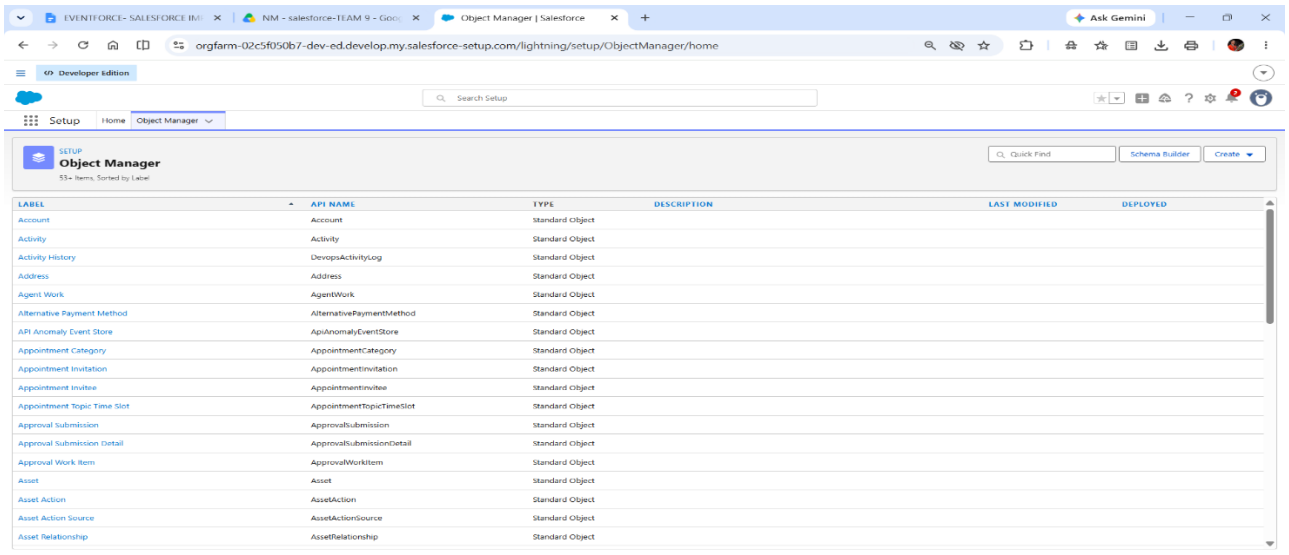
Event__c ➔ Client__c (Lookup) Event__c ➔

Venue__c (Lookup) Feedback__c ➔ Event__c

(Lookup) Feedback__c ➔ Client__c (Lookup)

EventVendor__c ➔ Event__c (Master-Detail) EventVendor__c

➔ Vendor__c (Master-Detail)



Formula Fields;

Event Budget Formula (Event_Budget_c - Currency): Standardizes dynamic budget estimation using a conditional CASE expression based on Event_Type__c:

sql

```
CASE(
  TEXT(Event_Type__c), "Wedding",
  50000,
  "Corporate", 30000,
  "Birthday", 10000,
  "Anniversary", 20000,
  "Festival", 60000,
  "Concert", 40000,
  "Other", 15000,
```

)

Feedback Contextual Lookup Filter: Applied on Feedback_c.Event_c so that selecting a Client restricts visible Events to records where Event_c.Client_c == Feedback__c.Client

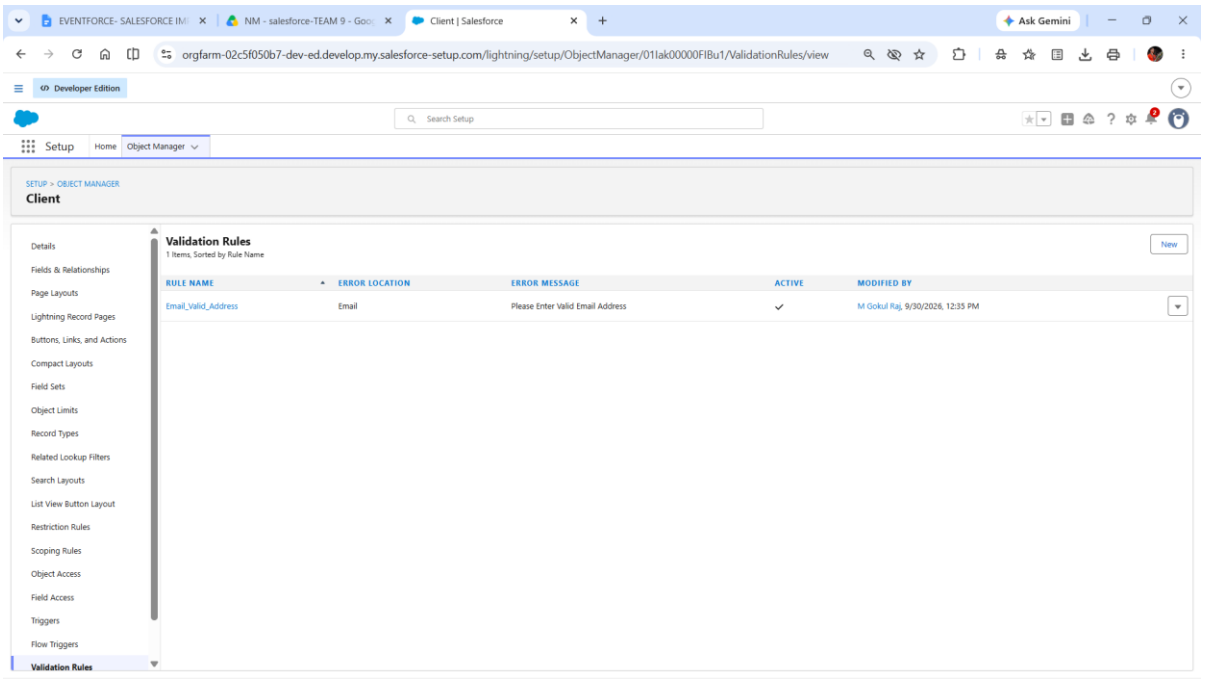
Validation Rules

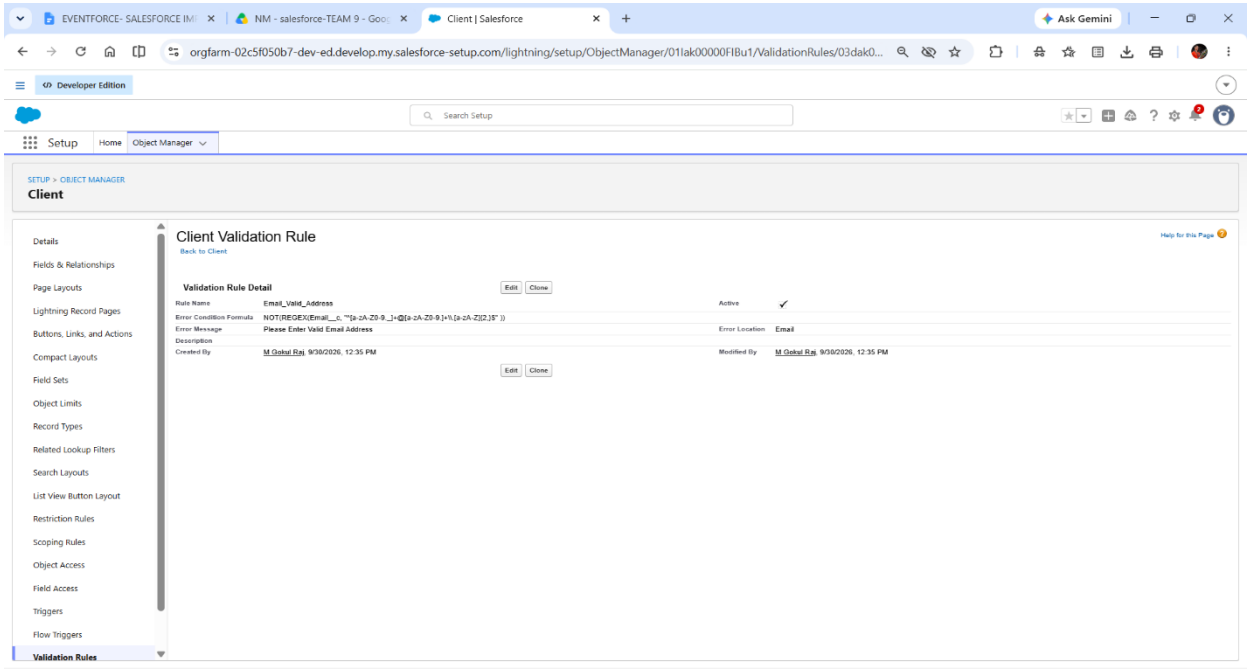
Email Format Enforcement: Implemented on `Client__c` using a regular expression to enforce valid email formats upon insert/update:

Rule Name: Email_Valid_Address

Formula: NOT(REGEX(Email__c, "[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\\.[a-zA-Z]{2,}$"))

Error Message:"Please Enter Valid Email Address"





Business Automation: Flows & Approval Processes

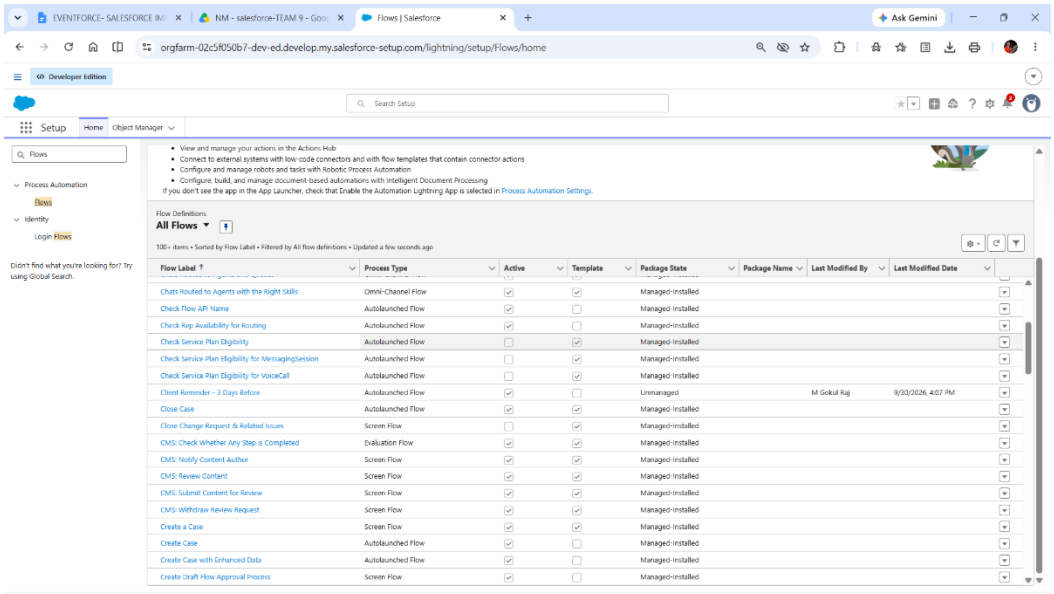
Record-Triggered Flow (Client Reminder - 3 Days Before):

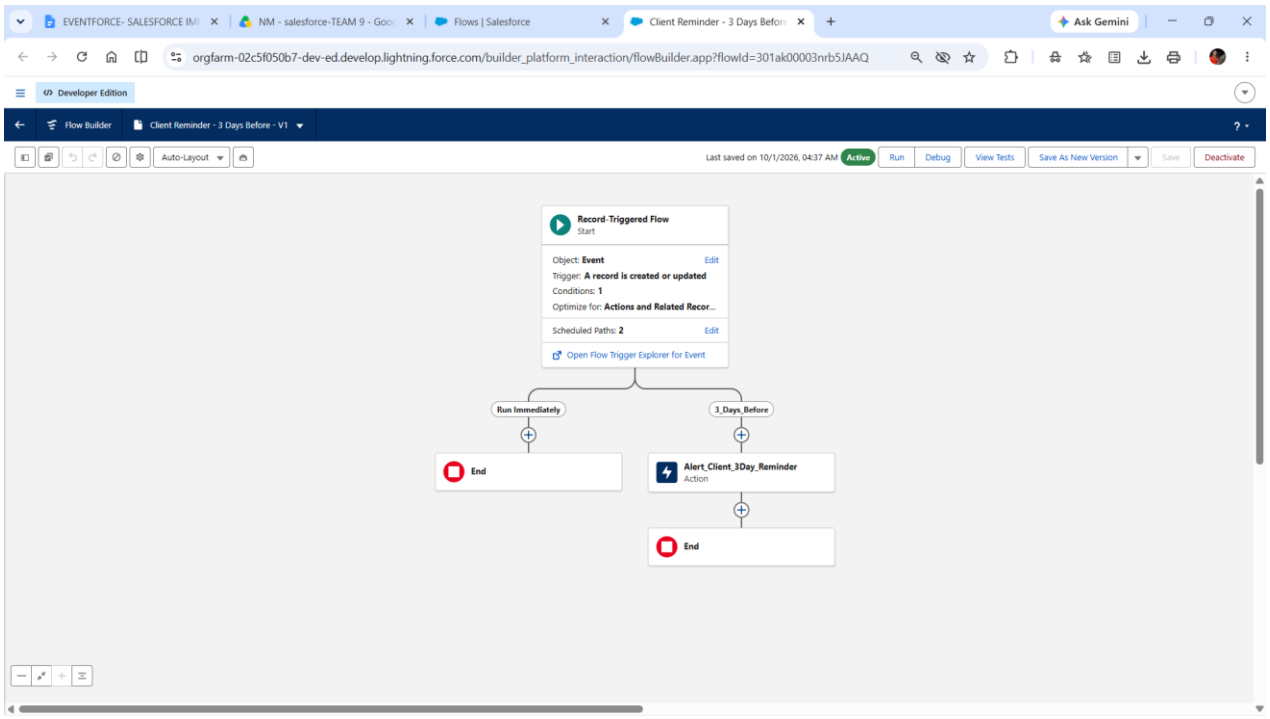
Trigger: Record created/updated with Event_Status__c = Confirmed.

Optimization: Actions and Related Records.

Scheduled Path: Configured for 3 Days Before the Event_Date__c field.

Action: Issues an automated email alert (Send Email) pulling plain-text merge templates to the assigned Client, CC-ing the Event Owner.





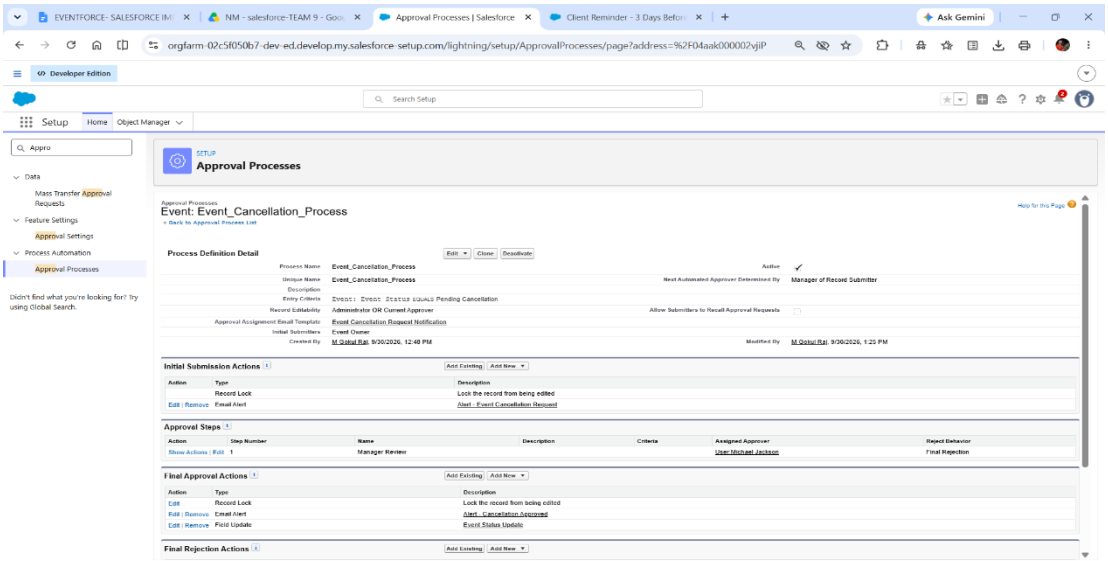
Event Cancellation Approval Process (Event_Cancellation_Process):

Entry Criteria: Event__c.Event_Status__c EQUALS "Pending Cancellation".

Submission Action: Locks the record and issues email alerts using Classic Email Templates to coordinators.

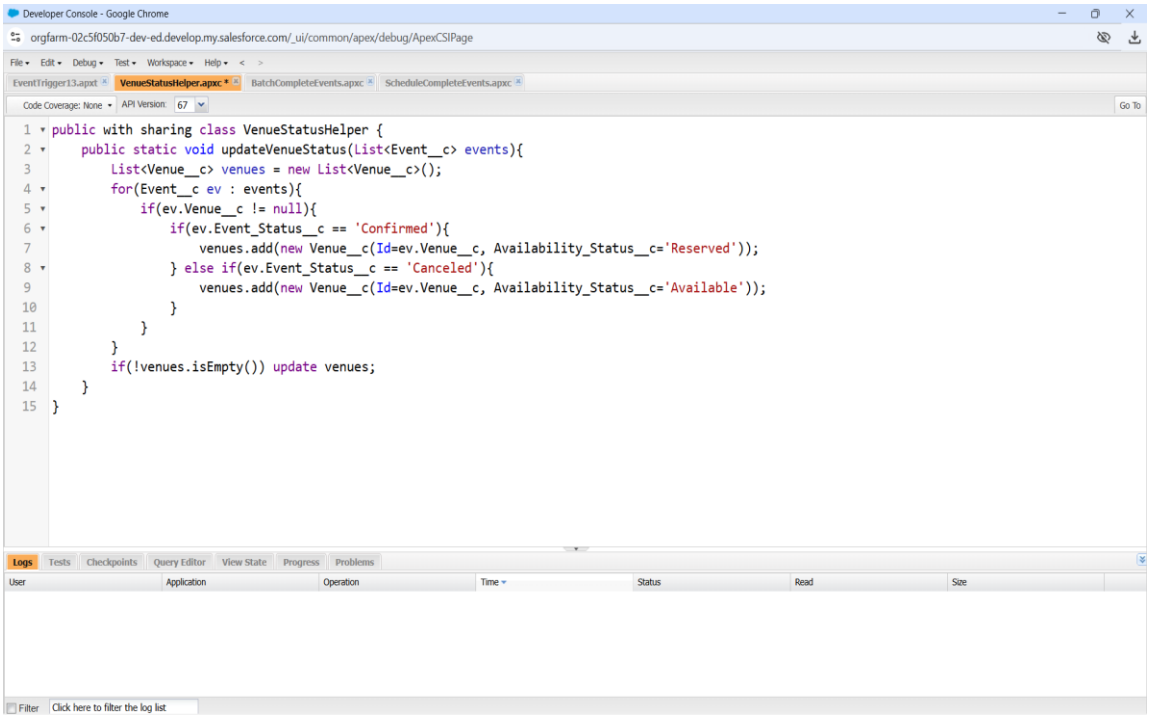
Approval Action: Updates Event_Status__c to Cancelled and notifies both the Event Owner and Client.

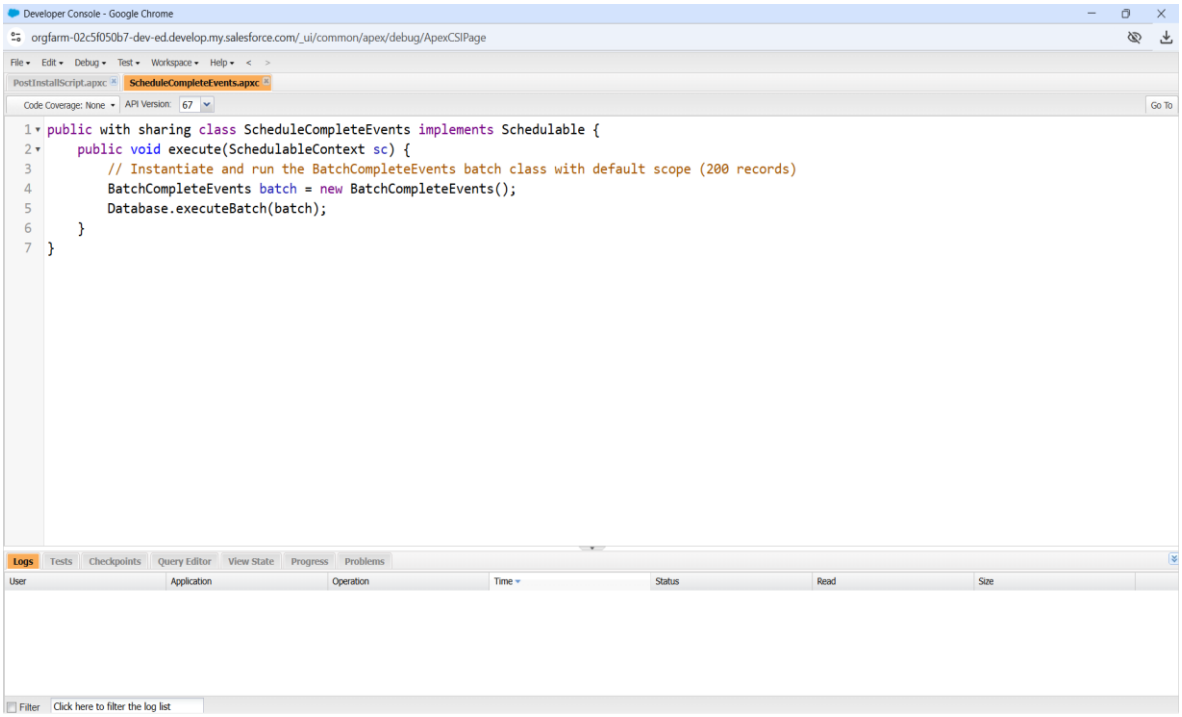
Rejection Action: Resets Event_Status__c to Rejected (or reverts to Confirmed) and alerts the Event Owner.



Apex Triggers & Scheduled Batch Jobs

Venue Status Automation: An after-insert/after-update trigger (EventTrigger13) invokes VenueStatusHelper.updateVenueStatus() to automatically toggle Venue_c.Availability_Status_c to Reserved when an event is confirmed, and reset it to Available if cancelled.





PROGRAM:

```
public with sharing class VenueStatusHelper {
    public static void updateVenueStatus(List<Event__c> events) { List<Venue_
        c> venues = new List<Venue__c>();

        for (Event__c ev : events) { if
            (ev.Venue__c != null) {
                if (ev.Event_Status__c == 'Confirmed') {
                    venues.add(new Venue__c(
                        Id = ev.Venue__c, Availability_Status_
                        c = 'Reserved'
                    ));
                } else if (ev.Event_Status__c == 'Canceled') {
                    venues.add(new Venue__c(
                        Id = ev.Venue__c, Availability_Status_
                        c = 'Available'
                    ));
                }
            }
        }

        if (!venues.isEmpty()) {
            update venues;
        }
    }
}
```

Double-Booking Prevention Trigger (PreventDoubleBooking): A before-insert/before-update trigger inspects existing database reservations and blocks updates if another event is already confirmed for that venue on the requested date:

PROGRAM:

```
trigger PreventDoubleBooking on Event__c (before insert, before update) { Set<Id>
   venueIds = new Set<Id>();

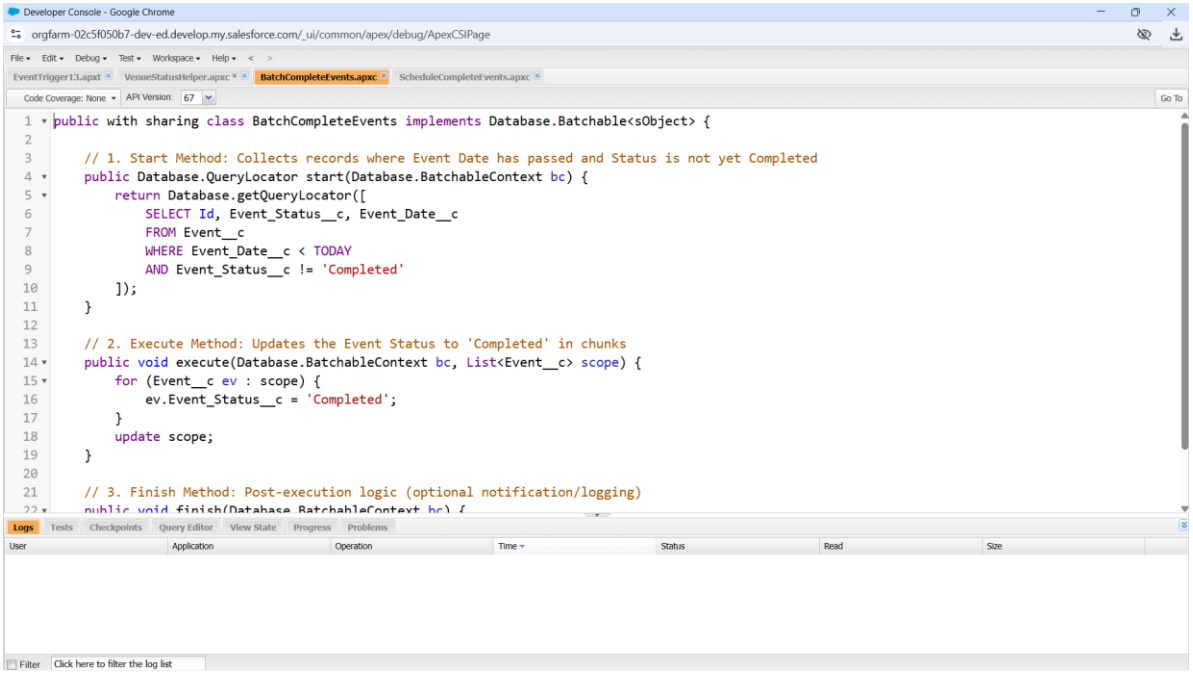
   for (Event__c ev : Trigger.new) { if
      (ev.Venue__c != null) {
         venueIds.add(ev.Venue__c);
      }
   }

   Map<Id, List<Event__c>> venueEventMap = new Map<Id, List<Event__c>>();
   for (Event__c e : [SELECT Id, Venue__c, Event_Date__c FROM
                  Event__c
                  WHERE Venue__c IN :venueIds]) {
      if (!venueEventMap.containsKey(e.Venue__c)) { venueEventMap.put(e.Venue
         c, new List<Event__c>());
      }
      venueEventMap.get(e.Venue__c).add(e);
   }

   for (Event__c ev : Trigger.new) {
      if (ev.Venue__c != null && venueEventMap.containsKey(ev.Venue__c)) {
         for (Event__c existing : venueEventMap.get(ev.Venue__c)) {
            if (existing.Event_Date__c == ev.Event_Date__c && existing.Id !=
ev.Id) {
               ev.addError('This Venue is already booked on this date.');
```

```
        }
    }
}
}
```

Daily Event Lifecycle Maintenance: A batch class (BatchCompleteEvents) queries past events (Event_Date_c < TODAY and status not completed) and updates them to Completed. This is invoked by a schedulable wrapper (ScheduleCompleteEvents) registered through the Salesforce Setup scheduler to run nightly at 8:00 PM.



28-C

PHASE 3: UI/UX CUSTOMIZATION & ANALYTICS

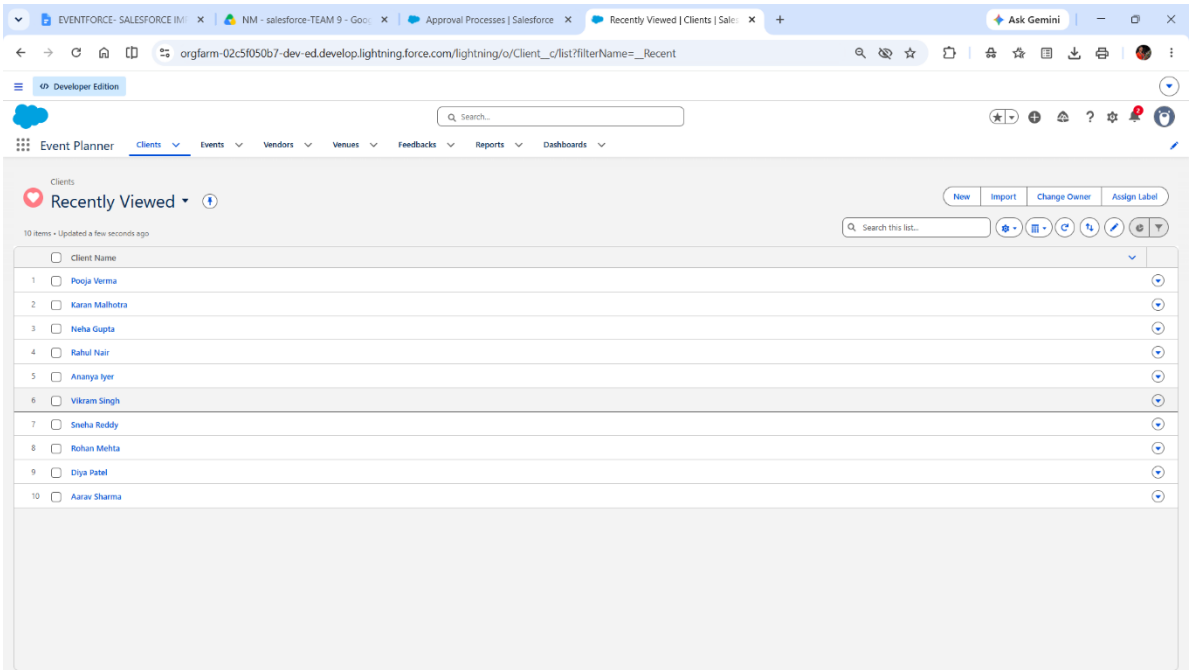
Custom Lightning App (Event Planner): Configured via Lightning App Manager with custom branding, an intuitive navigation bar, and accessible tabs: Events, Clients, Vendors, Venues, Feedbacks, Reports, and Dashboards.

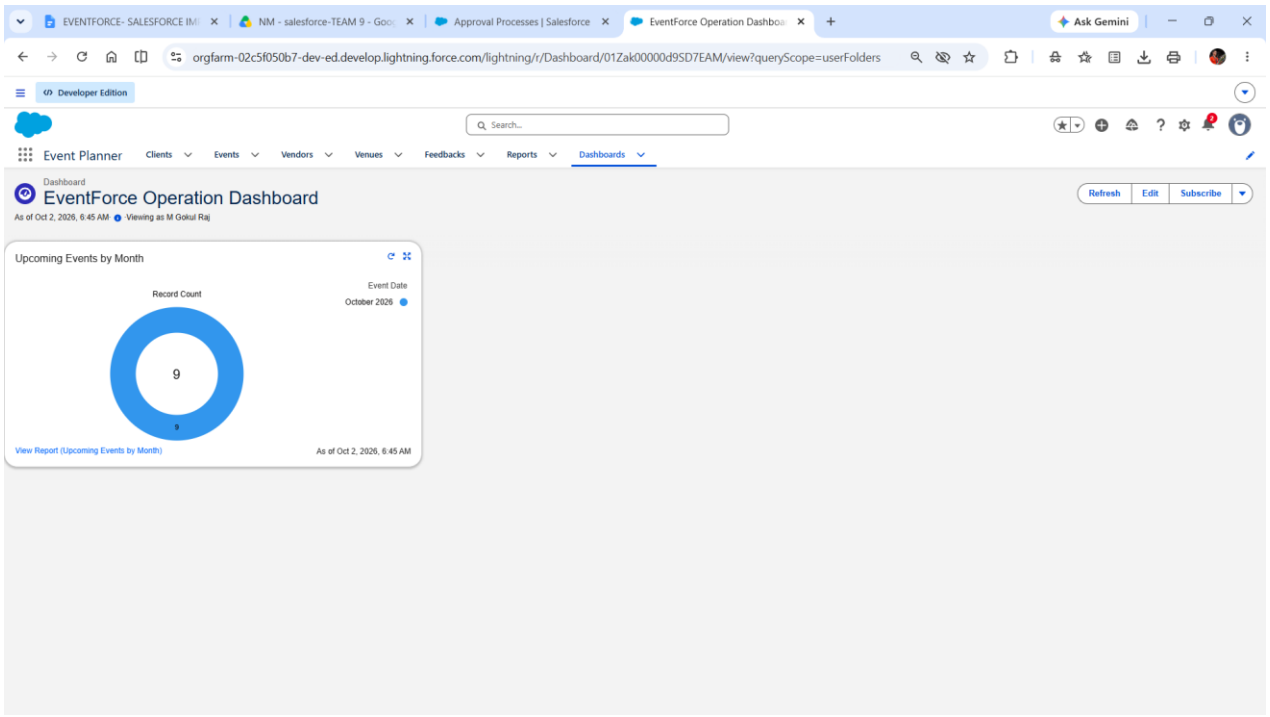
Operational Reporting:

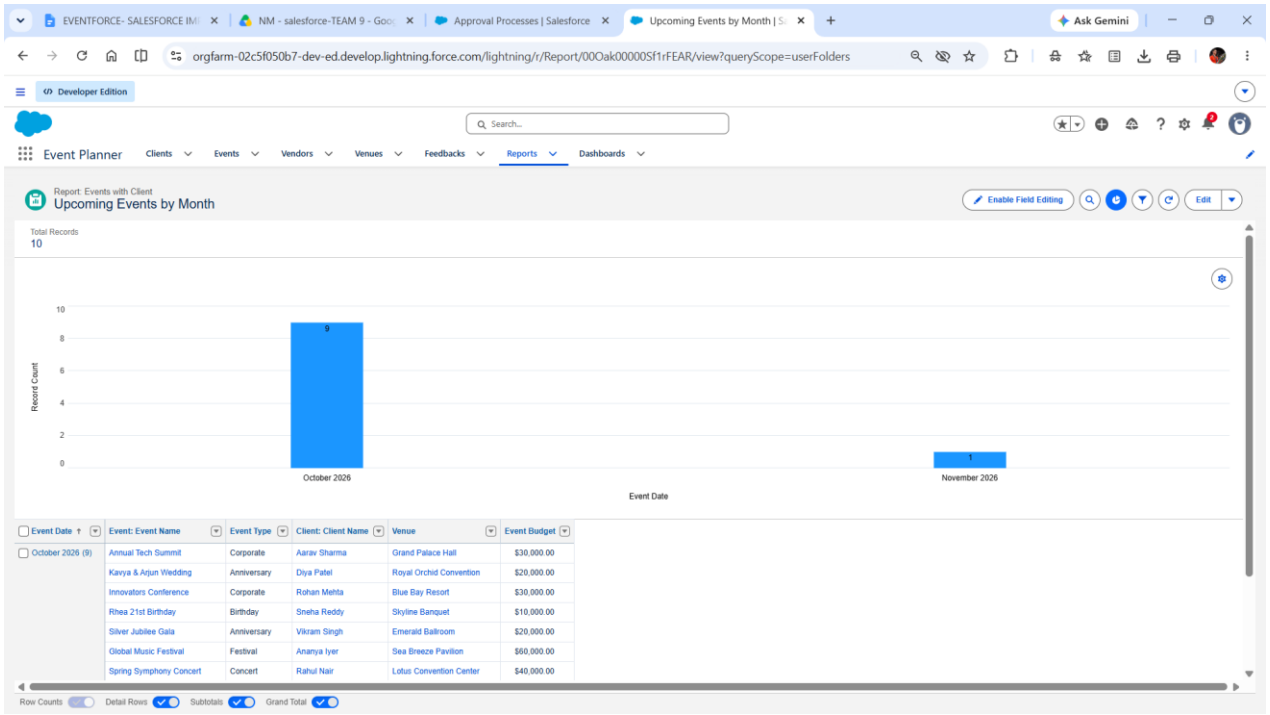
Report Folder: EventForce Report, shared with View access for the Event Coordinator and Vendor Manager roles.

Summary Report (Upcoming Events by Month): Uses grouping by calendar month on Event_Date__c, filters out canceled events, displays record counts, and sums total budget metrics (SUM(Event_Budget__c)).

Executive Dashboard (EventForce Operations Dashboard): Features donut and horizontal bar charts highlighting upcoming event volumes and cumulative revenue per calendar month.




EventForce Operation Dashboard
orgfarm-02c5f050b7-dev-ed.develop.lightning.force.com/lightning/r/Dashboard/01Zak00000d9SD7EAM/view?queryScope=userFolders
Developer Edition
Event Planner
Clients
Events
Vendors
Venues
Feedbacks
Reports
Dashboards
Dashboard
EventForce Operation Dashboard
As of Oct 2, 2026, 6:45 AM · Viewing as M Gokul Raj
Refresh
Edit
Subscribe
Upcoming Events by Month
Record Count
Event Date
October 2026
9
9
View Report (Upcoming Events by Month)
As of Oct 2, 2026, 6:45 AM


Upcoming Events by Month | S
orgfarm-02c5f050b7-dev-ed.develop.lightning.force.com/lightning/r/Report/00Oak00000Sf1rFEAR/view?queryScope=userFolders
Developer Edition
Event Planner
Clients
Events
Vendors
Venues
Feedbacks
Reports
Dashboards
Report: Events with Client
Upcoming Events by Month
Enable Field Editing
Edit
Total Records
10
Record Count
October 2026
November 2026
Event Date
Event Date
Event: Event Name
Event Type
Client: Client Name
Venue
Event Budget
October 2026 (9)
Annual Tech Summit
Corporate
Aarav Sharma
Grand Palace Hall
$30,000.00
Kavya & Arjun Wedding
Anniversary
Diya Patel
Royal Orchid Convention
$20,000.00
Innovators Conference
Corporate
Rohan Mehta
Blue Bay Resort
$30,000.00
Rhea 21st Birthday
Birthday
Sneha Reddy
Skyline Banquet
$10,000.00
Silver Jubilee Gala
Anniversary
Vikram Singh
Emerald Ballroom
$20,000.00
Global Music Festival
Festival
Ananya Iyer
Sea Breeze Pavilion
$60,000.00
Spring Symphony Concert
Concert
Rahul Nair
Lotus Convention Center
$40,000.00
Row Counts
Detail Rows
Subtotals
Grand Total

PHASE 4: DATA MIGRATION, TESTING & SECURITY SETUP

Data Migration: Baseline data for clients, venues, and vendors was cleaned, validated for duplicate constraints, and imported into Developer Edition using the Salesforce Data Import Wizard.

Organization-Wide Defaults (OWD):

Event__c: Set to Private (internal and external) to restrict baseline record visibility exclusively to record owners and higher-level roles.

Client__c, Vendor__c, Venue__c: Set to Public Read/Write.

Role Hierarchy: Scaled under executive leadership (CEO ➔ Event Admin ➔ child roles: Event Coordinator, Vendor Manager, Client) to mirror the corporate chain of command.

Profile-Level Access & Password Policies:

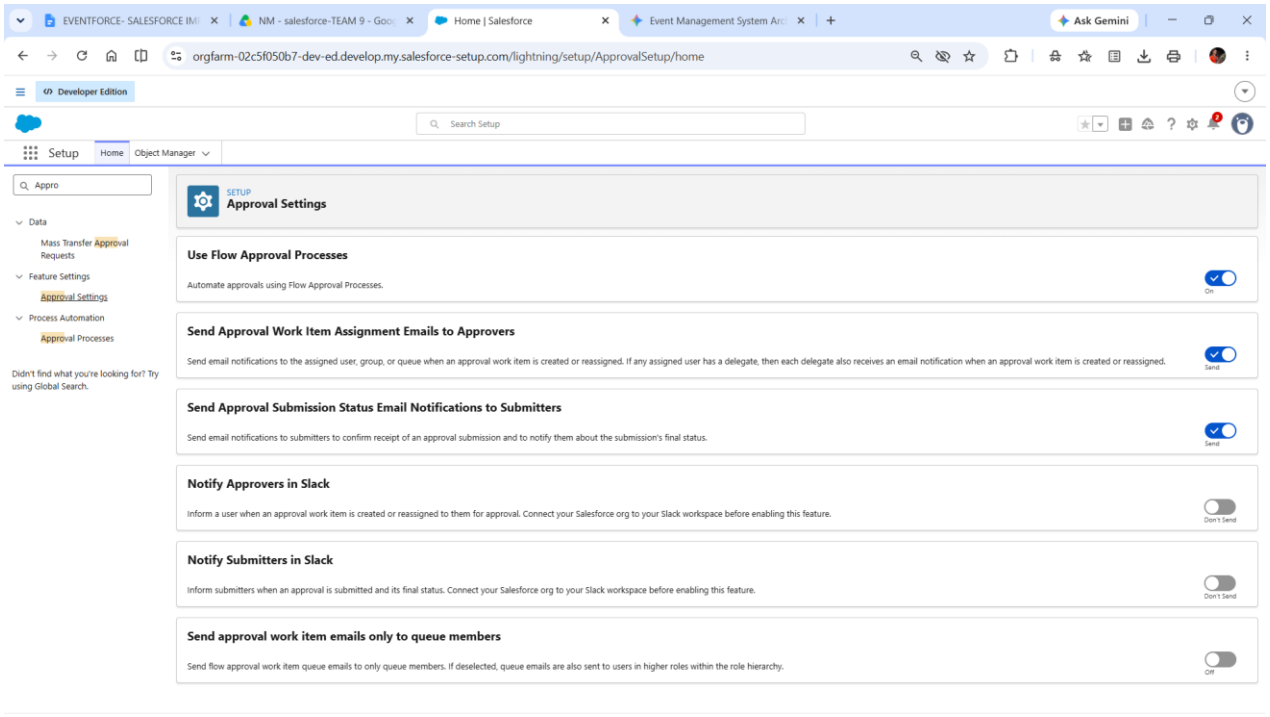
Event Admin: Cloned from System Administrator for full customization and data governance.

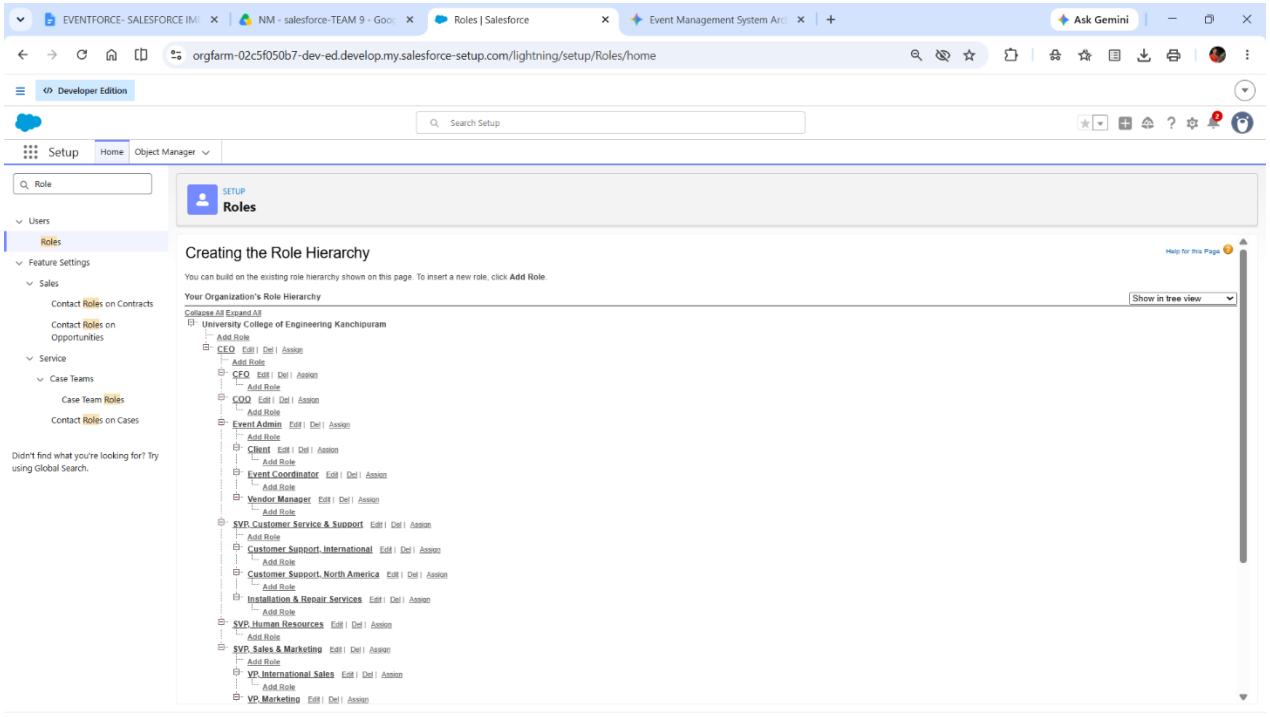
Event Coordinator: Base Standard Platform User with CRED on Events, Read/Edit on Clients, Read-Only on Vendors/Venues, and Create/Read on Feedback.

Vendor Manager: Base Standard Platform User with CRED on Vendors and Read-Only on Events, Venues, and Feedback.

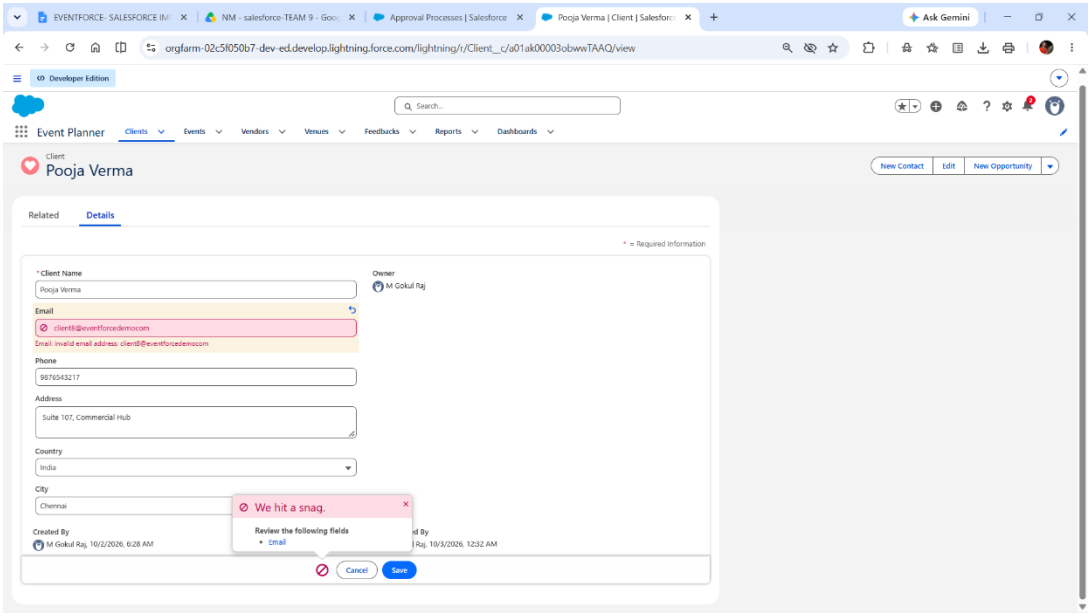
Client Profile: Restricted Read-Only visibility to personal Events and Feedback submission.

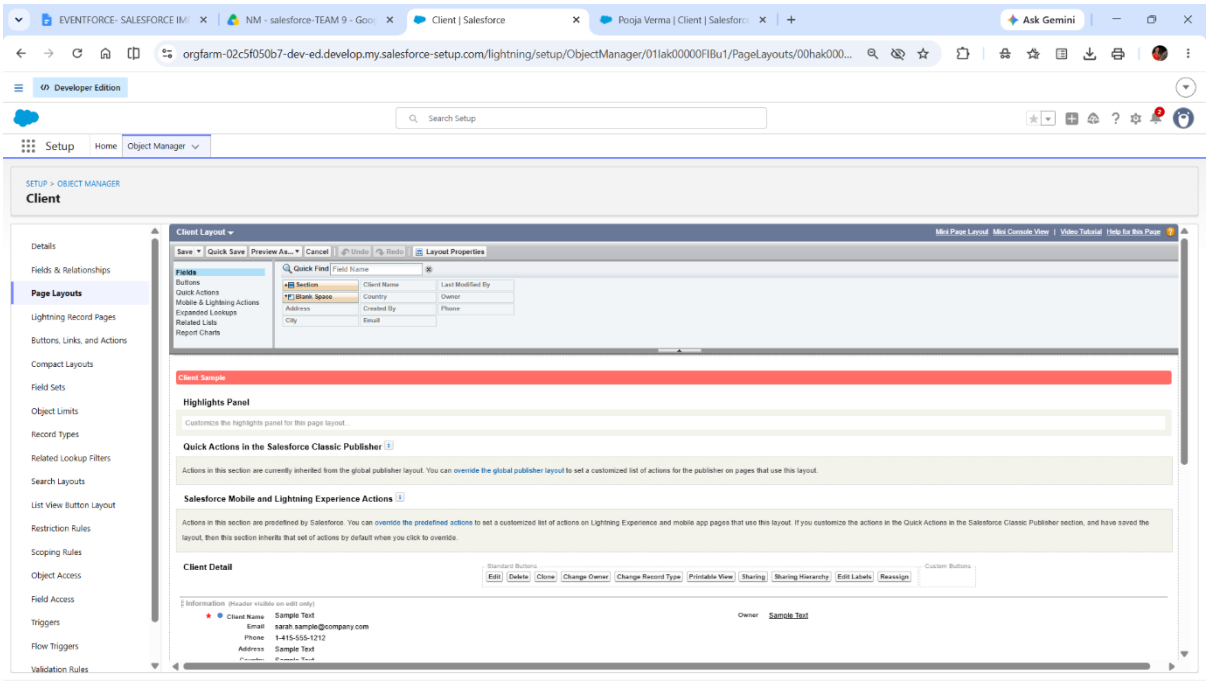
Permission Sets: Implemented the Feedback Manager permission set granting elevated Read/Edit/Create/Delete object rights to targeted coordinators without changing their base profile.


Home | Salesforce
orgfarm-02c5f050b7-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ApprovalSetup/home
Developer Edition
Search Setup
Setup
Home
Object Manager
Appro
Data
Mass Transfer Approval Requests
Feature Settings
Approval Settings
Process Automation
Approval Processes
Didn't find what you're looking for? Try using Global Search.
SETUP
Approval Settings
Use Flow Approval Processes
Automate approvals using Flow Approval Processes.
On
Send Approval Work Item Assignment Emails to Approvers
Send email notifications to the assigned user, group, or queue when an approval work item is created or reassigned. If any assigned user has a delegate, then each delegate also receives an email notification when an approval work item is created or reassigned.
Send
Send Approval Submission Status Email Notifications to Submitters
Send email notifications to submitters to confirm receipt of an approval submission and to notify them about the submission's final status.
Send
Notify Approvers in Slack
Inform a user when an approval work item is created or reassigned to them for approval. Connect your Salesforce org to your Slack workspace before enabling this feature.
Don't Send
Notify Submitters in Slack
Inform submitters when an approval is submitted and its final status. Connect your Salesforce org to your Slack workspace before enabling this feature.
Don't Send
Send approval work item emails only to queue members
Send flow approval work item queue emails to only queue members. If deselected, queue emails are also sent to users in higher roles within the role hierarchy.
Off


Roles | Salesforce
orgfarm-02c5f050b7-dev-ed.develop.my.salesforce-setup.com/lightning/setup/Roles/home
Developer Edition
Search Setup
Setup
Home
Object Manager
Role
Users
Roles
Feature Settings
Sales
Contact Roles on Contracts
Contact Roles on Opportunities
Service
Case Teams
Case Team Roles
Contact Roles on Cases
Didn't find what you're looking for? Try using Global Search.
SETUP
Roles
Creating the Role Hierarchy
Help for this Page
You can build on the existing role hierarchy shown on this page. To insert a new role, click Add Role.
Your Organization's Role Hierarchy
Show in tree view
Collapse All Expand All
University College of Engineering Kanchipuram
Add Role
CEO Edit | Del | Assign
Add Role
CFO Edit | Del | Assign
Add Role
COO Edit | Del | Assign
Add Role
Event Admin Edit | Del | Assign
Add Role
Client Edit | Del | Assign
Add Role
Event Coordinator Edit | Del | Assign
Add Role
Vendor Manager Edit | Del | Assign
Add Role
SVP, Customer Service & Support Edit | Del | Assign
Add Role
Customer Support, International Edit | Del | Assign
Add Role
Customer Support, North America Edit | Del | Assign
Add Role
Installation & Repair Services Edit | Del | Assign
Add Role
SVP, Human Resources Edit | Del | Assign
Add Role
SVP, Sales & Marketing Edit | Del | Assign
Add Role
VP, International Sales Edit | Del | Assign
Add Role
VP, Marketing Edit | Del | Assign

Criteria & Sharing Rules: Deployed rule Event_Sharing_For_Vendors, automatically granting Read Only access to events owned by members of the Event Coordinator role to users assigned to the Vendor Manager role





PHASE 5: DEPLOYMENT STRATEGY, MAINTENANCE & DOCUMENTATION

Deployment Strategy

In this project lifecycle, all configurations, custom metadata, and code components were implemented and validated inside a standalone Salesforce Developer Edition Org. While academic and sandbox developer orgs operate independently of production environments, enterprise implementations utilize the following deployment strategy:

Environment Architecture: Development ➔Sandbox / Scratch Org ➔UAT (User Acceptance Testing) ➔Staging ➔Production.

Migration Mechanism: Deployment of metadata packages (Custom Objects, Layouts, Flows, Triggers, Profiles) using Salesforce CLI (SFDX), change sets, or Git-based CI/CD pipelines (e.g., GitHub Actions / Jenkins).

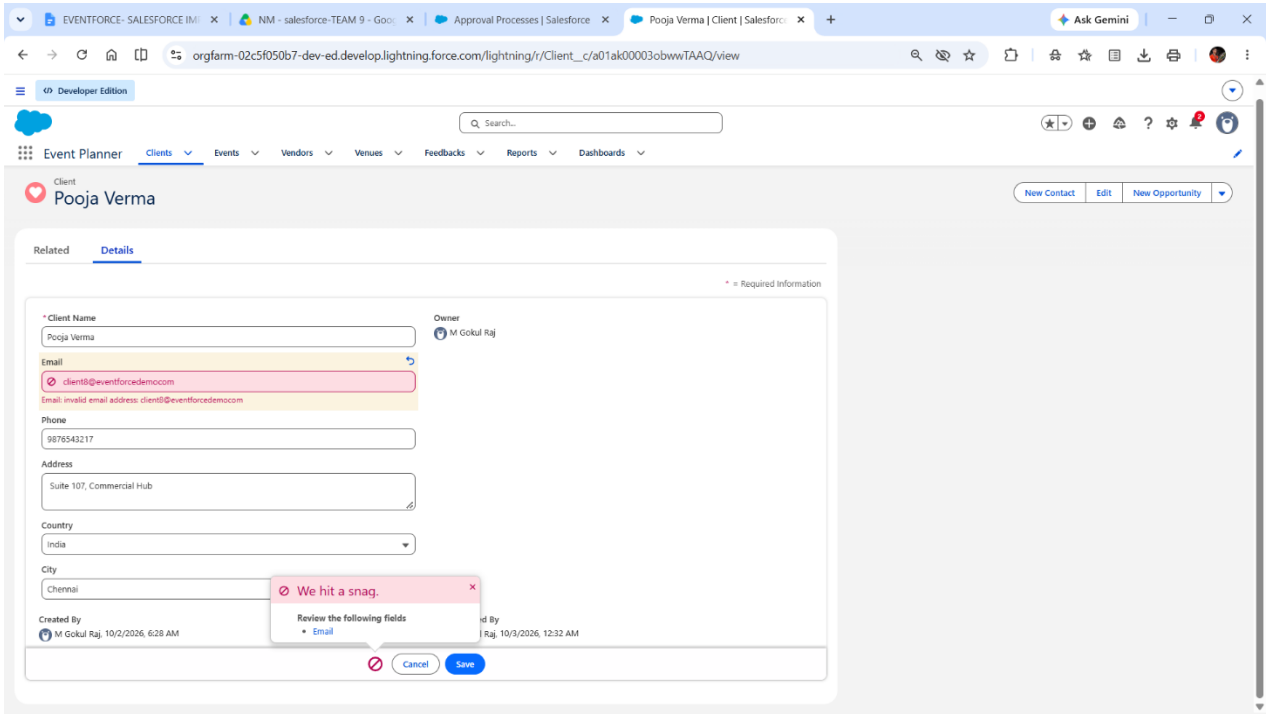
Pre-Deployment Checklist: Execute all test methods to confirm an Apex unit test coverage above the mandatory 75% threshold, validate Flow trigger ordering, and ensure profile/permission set parity.

Maintenance, Monitoring & Troubleshooting

Automated Log Monitoring: Track apex execution failures, flow run errors, and email delivery exceptions via Setup Apex Jobs, Paused and Failed Flow Interviews, and Email Logs.

Data Hygiene Maintenance: Run weekly duplicate check routines and verify that the nightly `ScheduleCompleteEvents` scheduled batch job consistently updates past event records.

Access Control Audits: Regularly audit profile assignments, permission sets, and sharing rules to maintain least-privilege access and avoid privilege escalation.



Documentation Summary

The project documentation serves as the single source of truth for all stakeholders, administrators, and developers. It captures the underlying business rationale, full data dictionary, entity-relationship models, process automation logic, role-based access models, and operational runbooks for continued platform evolution.

Conclusion & Project Outcomes

The EventForce Management System implementation highlights the power of Salesforce as an extensible, enterprise-grade application engine. By balancing declarative tools (Flows, Approvals, Page Layouts) with programmatic customizations (Triggers, Schedulable Apex), the platform achieves end-to-end automation while maintaining strict security via fine-grained OWD and role hierarchies.

The resulting CRM successfully consolidates event booking operations, eliminates manual scheduling conflicts, provides real-time budget forecasting, and ensures a seamless experience for clients, coordinators, and vendors alike.